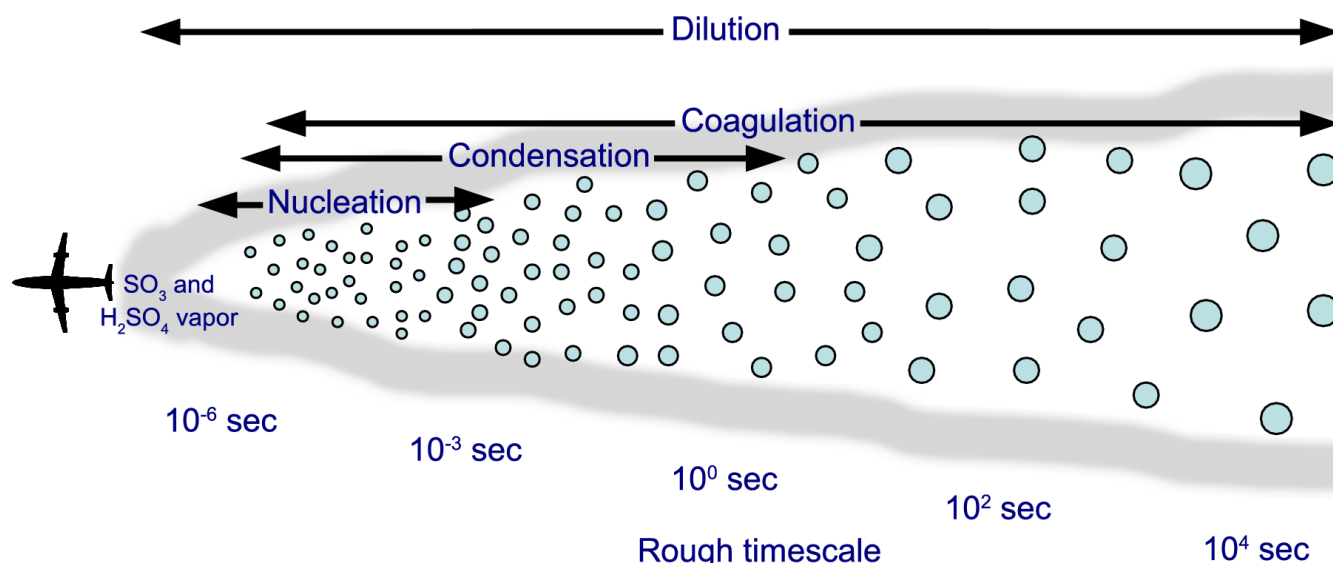


1 **Auxiliary Material for “Efficient formation of stratospheric aerosol for climate**
2 **engineering by emission of condensible vapor from aircraft” by Jeffrey R. Pierce,**
3 **Debra K. Weisenstein, Patricia Heckendorn, Thomas Peter, and David W. Keith**



5 Figure s1. Schematic showing the rough timescales where various processes are important within the
6 aircraft plume following H_2SO_4 emission. The timescales for the processes vary from the values shown
7 here depending on the nucleation, condensation and dilution assumptions tested in the plume model.

8
9 **Aircraft plume aerosol model**

10 To model aerosol formation and growth within the aircraft plume, we use an expanding
11 Lagrangian-box version of the Two-Moment Aerosol Sectional (TOMAS) microphysics model [Adams
12 and Seinfeld, 2002]. The box tracks a radial cross section of the plume with time, assuming that the
13 plume is well mixed radially and dilutes with time [Yu and Turco, 1998]. This version of TOMAS
14 [Pierce and Adams, 2009] simulates aerosol number and mass in 43 lognormally spaced size bins with
15 diameters ranging from 0.6 nm to 10 μm . For all simulations of the initial plume, we assume that any
16 emitted SO_3 (see injection methods for discussion) has already reacted to form H_2SO_4 (as justified by

17 the large rate coefficient of $\text{SO}_3 + \text{H}_2\text{O}$ under the moist aircraft-plume conditions [*JPL*, 2006]).
18 Therefore, the model is initialized with only H_2SO_4 vapor and background aerosols at a temperature of
19 220 K and a relative humidity of 10% (the results were not sensitive to reasonable perturbations in
20 these inputs). The concentration of the aqueous sulfuric acid solutions in the aerosol droplets is
21 assumed to be in equilibrium with vapor-phase water.

22 We explore several uncertain assumptions within the plume (Table s1). The first is the plume
23 dilution rate. There is limited information on the dilution rates in stratospheric jet plumes; therefore,
24 these represent an uncertain aspect of this study. To test some potential variability in the plumes, we
25 define slow and fast dilution rates that are similar to those of *Yu and Turco* [1998] describing the plume
26 dispersion during the first 16 minutes after injection. After this time, the fast plume then dilutes at the
27 same rate as the slow plume. The equations defining the cross-sectional area of the plume as a function
28 of time are given in Table s1. The second uncertain assumption is the nucleation mechanism and rate.
29 To address this we scale the kinetic barrierless nucleation theory of *Clement and Ford* [1999] (every
30 pair of H_2SO_4 molecules that collide form a stable cluster, an upper limit of nucleation rate). We chose
31 to use barrierless nucleation because (1) nucleation should approach the barrierless limit for very high
32 H_2SO_4 vapor concentrations and low temperatures (initial H_2SO_4 concentrations in the aircraft plume
33 are on the order of $10^{15} \text{ molec cm}^{-3}$, about six orders of magnitude higher than the highest H_2SO_4
34 concentrations in the atmosphere, and the temperature is 220 K) and (2) there are no other available
35 nucleation parameterizations that can handle these high concentrations of sulfuric acid. To address the
36 fact that the nucleation may not actually be as fast as barrierless nucleation would predict, we scale the
37 rates by four scaling factors ranging from 1 to 10^{-9} . The third uncertain assumption is the H_2SO_4
38 condensation accommodation coefficient (sticking probability). Values in the literature have been
39 shown to range [*Seinfeld and Pandis*, 2006] from 0.01 to 1 and we test three values within this range.
40 The fourth assumption is the sulfur emission rate per length of aircraft flight path, and we test 3 and 30

41 kg S km⁻¹ of flight. H₂SO₄ emission rates per unit length along the flight track were chosen to yield
42 size distributions that roughly bracket the radii that have the most favorable optical properties (Figure
43 1). These rates correspond to an aircraft delivering between 3 and 30 tonnes of sulfur over a 1000 km
44 flight in the stratosphere. The final uncertain assumption is the number concentration of background
45 stratospheric particles. We assume either 1 or 50 cm⁻³ to cover pristine and geoengineered stratospheric
46 conditions (see Figure s3), both in a single lognormal mode with a number-median radius of 0.2 μm
47 and a width parameter of 2.0 (see Figure 3). The background aerosol makes little difference to the
48 resultant size distributions in the plume because 1) it represents a trivial fraction of the condensation
49 sink when H₂SO₄ is condensing and 2) it does not dominate the coagulation sink for plume particles
50 until the point where the plume particles are passed to the global model. We do not test uncertainties in
51 coagulation since uncertainties in Brownian coagulation are generally much smaller than the
52 uncertainties explored above [Seinfeld and Pandis, 2006].

53 As discussed in the main article, the final size distribution reached by the plume model (at the
54 time when the coagulation of the plume particles with pre-existing stratospheric particles dominates
55 over self-coagulation of the plume particles, which happens after several days for the geoengineered
56 atmosphere and longer in the pristine) depends almost entirely on the sulfur emission rate per unit
57 aircraft flight path length and the plume dispersion rate. The results were highly insensitive to the
58 nucleation scale factors and the accommodation coefficients due to nucleation and condensation only
59 occurring during the initial seconds to minutes of the plume. Beyond these initial seconds to minutes,
60 the aerosol microphysics is dominated by self-coagulation of particles formed in the plume.
61 Coagulation is proportional to the square of the particle concentration, so any initial differences in the
62 particle size distribution would be damped by coagulation. These results are consistent with *Turco and*
63 *Yu* [1999].

64 The smoothing of initial differences in the size distribution is illustrated in Figure s2. Each

65 panel shows the results of 12 simulations of the various combinations of nucleation scaling factors and
66 accommodation coefficients. Figure s2a shows the aerosol size distributions at 1 ms into the fast-
67 diluting plume with an ambient stratospheric aerosol concentration of 50 cm^{-3} at the emission rate of 3
68 kg km^{-1} . Here, very large differences between the simulations are apparent because of the large
69 differences in nucleation and condensation rates. Figures s2b and s2c show the aerosol number size
70 distribution 1 hour after injection for the slow- and fast-diluting plumes, respectively, with an ambient
71 stratospheric aerosol concentration of 50 cm^{-3} at the emission rate of 3 kg km^{-1} . After 1 hour in the
72 slowly diluting plume, there is very little difference between the simulations with different nucleation
73 scale factors and accommodation coefficients. In the fast-diluting plume, only the slowest nucleation
74 cases tend to deviate slightly from the others. Thus, the aerosol size distributions show little sensitivity
75 to the accommodation coefficient and nucleation rate, which allows us to ignore exact details of the
76 nucleation mechanisms (e.g. the role of ions [Yu and Turco, 1998]) and the accommodation coefficient.

77 There are several other uncertain factors that we do not explicitly address in this analysis. (1)
78 The amount of primary soot produced by the aircraft is uncertain; however Yu and Turco [1998] found
79 that it made little difference on the nucleation and growth of particles in conventional jet plumes in the
80 stratosphere. For our cases with additional H_2SO_4 emissions, the role of soot particles will be even
81 smaller. (2) We assume that SO_3 combines with H_2O to form H_2SO_4 instantly upon cooling [Yu and
82 Turco, 1998; JPL, 2006]. We discuss this issue further in the section on injection methods later in this
83 supplementary material. (3) We assume that coagulation rates are accurate. Although charged particles
84 can alter the coagulation rates of the smallest particles [Yu and Turco, 1998], Brownian coagulation
85 rates are generally much more certain than nucleation and condensation rates [Seinfeld and Pandis,
86 2006]. (4) We assume that the plume is well mixed radially. In reality, this is not the case, and the rates
87 of microphysical processes will vary radially (and temporally in the turbulent areas of plume near the
88 aircraft). Spatial and temporal inhomogeneities in the plume would likely broaden the size distribution

89 of emitted particles. This must be explored in future work. (5) We assume that the emissions always
 90 occur in areas away from other recently emitted plumes (within past ~2 days) such that the interaction
 91 between plumes is insignificant. The validity of this assumption will depend on aircraft flight
 92 strategies that are out of the scope of this paper.

93

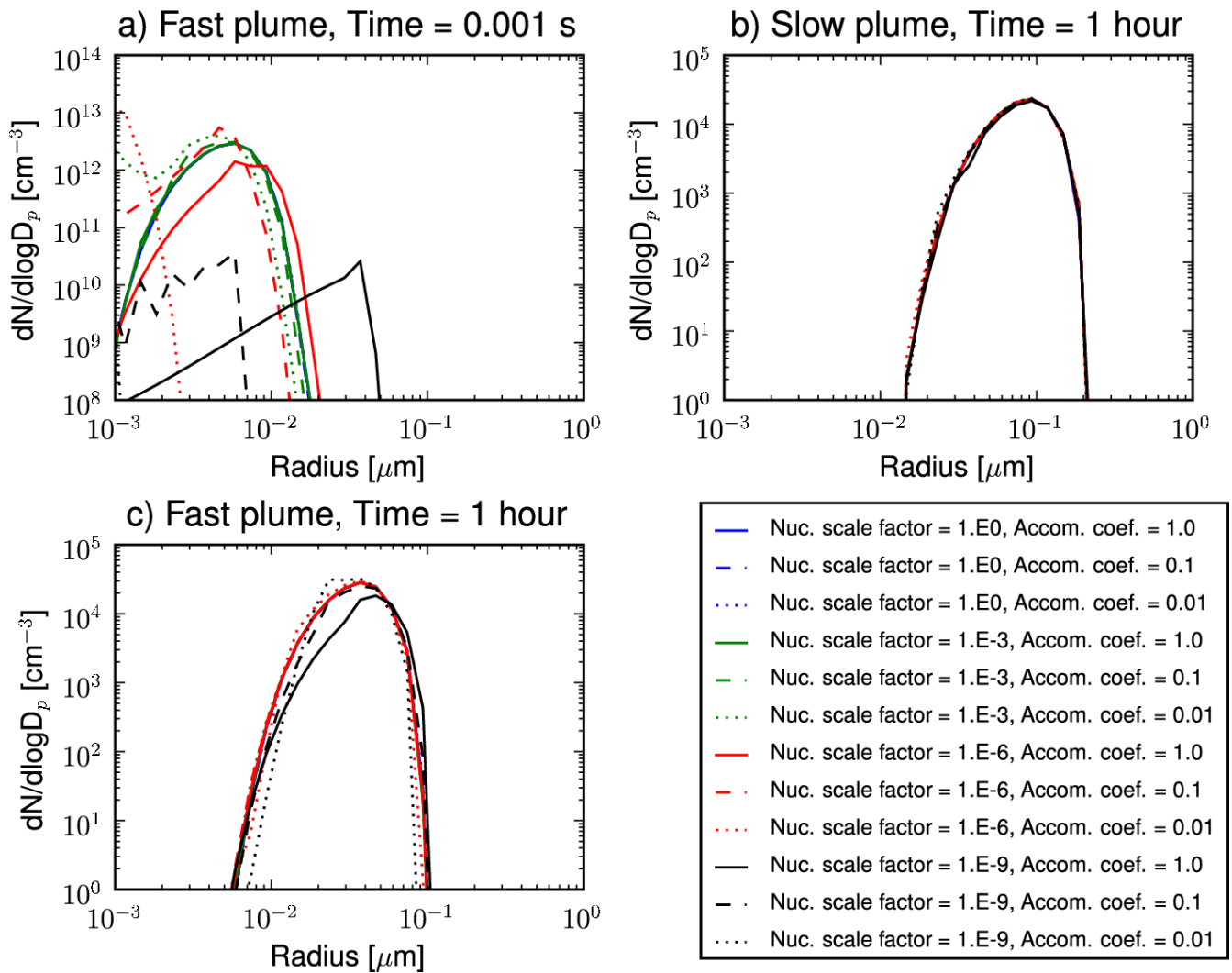
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95 Table s1: Uncertain parameters tested in plume model.

	Values tested
Plume x-sect. area [m²]	Slow = $6+10^{\log(t[s])+2}$, Fast = $10+10^{\log(t[s])+3}$
Nucleation scale factor	1, 10^{-3} , 10^{-6} , 10^{-9}
Accommodation coefficient	1, 0.1, 0.01
Sulfur injection rate [kg km⁻¹]	3, 30
Background number concentration [cm⁻³]	1, 50

96

97



98

99 Figure s2. Particle number size distributions a) after 0.001 second in the fast-diluting plume, b) after 1
 100 hour in the slow-diluting plume, and c) after 1 hour in the fast-diluting plume. Twelve simulations
 101 represent all combinations of the four nucleation scale factor assumptions (represented by colors) and
 102 three accommodation coefficient assumptions (represented by line type).

103

104

105 Stratospheric aerosol model

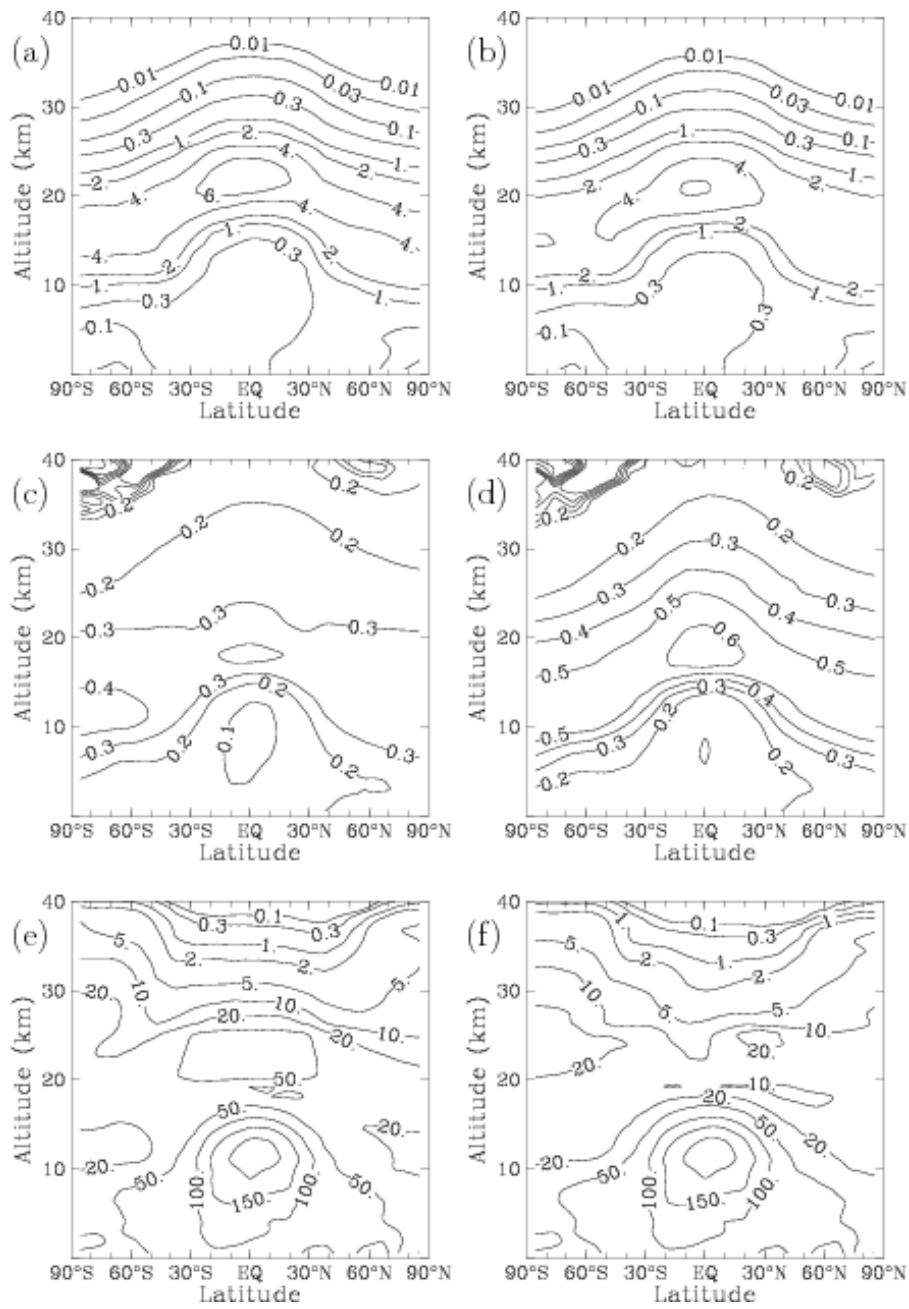
106 To determine how the aerosol burden and radiative forcing differ between SO₂- and H₂SO₄-

107 injection schemes, we use the AER 2-D aerosol model [*Heckendorn et al.*, 2009; *Weisenstein et al.*,
108 1997; *Weisenstein et al.*, 2007]. This model resolves latitude and altitude (resolution approximately
109 9.5° and 1.2 km, respectively), and it assumes that the atmosphere is homogeneous longitudinally. The
110 model domain is global from the surface to 60 km altitude. The assumption of zonally homogeneous
111 conditions is an acceptable approximation in the stratosphere a few kilometers away from the
112 tropopause [*Weisenstein et al.*, 2007]. The configuration of the model is the same as in *Heckendorn et*
113 *al.* [2009] with 40 size bins spanning the range 0.4 nm to 3.2 μm by volume doubling. Only pure
114 $\text{H}_2\text{SO}_4/\text{H}_2\text{O}$ aerosols (often called “sulfate aerosols”) are included, with surface emissions of OCS and
115 SO_2 providing the stratospheric sulfur input during volcanically quiescent periods; DMS, CS_2 , and H_2S
116 emissions are also included but primarily influence the model's troposphere. Microphysical processes
117 include homogeneous nucleation, condensation, coagulation, evaporation, and gravitational
118 sedimentation. Above about 35 km aerosols evaporate and photolysis of H_2SO_4 produces SO_2 gas
119 which later descends in the polar regions, reacts with OH, and nucleates and condenses to form new
120 particles.

121 Geoengineering emissions are input to the 2-D model as particles, using the final size
122 distributions from the plume model (number-median dry radius of 65, 95, or 180 nm, sigma of 1.5)
123 after adjusting dry radius to wet radius. The input particle number density is scaled to match the
124 desired total sulfur input and spread over the 30°N-30°S, 20-25 km region (in *Heckendorn et al.*
125 [2009], the emissions were all at the equator and 20 km). We assume here that individual plumes do
126 not cross within two days. The model is integrated with continuous emissions until the aerosol
127 concentrations in the stratosphere reach an annually-repeating steady state.

128 The calculated annual-average aerosol mass density, effective radius, and number density from
129 the 2-D model for geoengineering cases with 5 MT S yr^{-1} of emission of H_2SO_4 -derived 95 nm particles
130 and SO_2 with emissions spread between 30°S and 30°N and 20 and 25 km are shown in Figure s3. This

131 figure shows that for a constant sulfur emissions rate, there is a higher mass density of sulfate and a
132 smaller effective radius for the H₂SO₄-injection cases throughout the stratosphere. The fractional
133 reduction in the effective radius is larger than the fractional increase in mass density. Since the
134 scattering efficiency (Figure 1) is very sensitive to the effective radius at these sizes, a lower mass
135 sulfate burden would be required to achieve a given radiative cooling using the H₂SO₄-injection
136 scenarios. Number densities shown in Figure s3 are for particles with radius greater than 0.01 μm ,
137 which represents all particles outside nucleation regions. The SO₂ emission cases lead to nucleation
138 and a very broad size distribution (see Figure 3) which includes also more large particles and thus a
139 larger effective radius, but a smaller number of particles greater than 0.01 μm . The H₂SO₄-emission
140 calculation shown here is used as the ambient background for the plume simulations, with number
141 concentration of 50 cm^{-3} in the injection region, as seen in Figure s3e.



142

143 Figure s3. Calculated annual-average aerosol mass density [$\mu\text{g m}^{-3}$] (a and b), effective radius [μm] (c
 144 and d), and number density of particles with radius greater than 0.01 μm [cm^{-3}] (e and f) calculated by
 145 the 2-D model for geoengineering cases with 5 MT S yr⁻¹ emitted as particles with mean radius of 95
 146 nm (a, c, and e) and 5 MT S yr⁻¹ emitted as SO₂ (b, d and f) with emissions spread between 30°S and
 147 30°N and 20 and 25 km.

148

149 **Injection location**

150 In the article we focus on sulfur injection evenly distributed between 30°N and 30°S and
151 between 20 and 25 km; however, in Figures 3 and 4 we also include cases where SO₂ is injected only at
152 the equator and 20 km (for comparison to *Heckendorn et al.* [2009]). Broadening the injection location
153 latitudinally and vertically increases both the burden and the radiative cooling for a given injection rate
154 of SO₂. This effect is largely due to a reduction in the coagulation of particles because of the more
155 dilute injection scenario. It is clear from Figure 4 that the injection location is very important and must
156 be explored systematically in future work.

157

158 **Radiative forcing calculation**

159 Because our goal is to compare the flux changes in top-of-atmosphere (TOA) shortwave (SW)
160 radiation caused by various aerosol distributions, we calculate the radiative flux change directly from
161 the aerosol scattering properties that are, in turn, computed using Mie theory. The solar-band radiative
162 flux change is computed using a standard approximation [*Seinfeld and Pandis*, 2006]:

$$163 \quad \Delta F = - \sum_{s=1}^{40} \sum_{l=1}^{19} \bar{F}_l T_a^2 (1-A)(1-R)^2 \bar{\beta}_{s,l}^- \tau_{s,l} f_l$$

164 where ΔF is the change in the top-of-atmosphere shortwave flux, s is the index of 40 size bins and l is
165 the index of the 19 latitudes in the AER model, $\bar{F}_l$ is the latitudinally dependent, annually and
166 diurnally averaged shortwave flux at top-of-atmosphere, T_a is the fractional transmission of shortwave
167 radiation to the scattering layer, A is the fraction of area covered by clouds, R is the surface albedo,
168 $\bar{\beta}_{s,l}^-$ is the aerosol-size dependent and latitudinally dependent, annually and diurnally averaged aerosol
169 upscatter fraction, $\tau_{s,l}$ is the aerosol-size dependent and latitudinally dependent optical depth of the
170 stratospheric aerosols, and f_l is the fraction of the Earth's area covered by latitude section, l .

171 For simplicity we assume that the aerosol is purely scattering (i.e. neglect absorption) and exists

172 as a single layer entirely above clouds. We ignore multiple scattering. We assume T_a is exactly 1 at
 173 the stratospheric scattering layer. For A and R , we use globally averaged values of 0.6 and 0.15,
 174 respectively [Seinfeld and Pandis, 2006]. The instantaneous β values (prior to diurnal and annual
 175 averaging) are calculated from Mie theory assuming a refractive index of 1.4 and averaging over the
 176 solar spectrum.

177 The changes in TOA SW fluxes shown in our paper differ from those in *Heckendorn et al.*
 178 [2009] in that they do not include feedbacks in temperature, dynamics, or trace gases induced by
 179 geoengineering. However, we have compared the TOA SW flux changes predicted by this simple
 180 scheme to online radiative transfer calculations of the global Chemistry Climate Model (CCM) SOLar
 181 Climate Ozone Links Version 2.0 (SOCOLv2.0), the same model used for SW flux calculations in
 182 *Heckendorn et al.* [2009]. As in *Heckendorn et al.* [2009], Socol is run for 20 years to allow the
 183 atmosphere to adjust to a new steady state. Table s2 shows a comparison of TOA SW radiative flux
 184 changes predicted by the simple model and Socol. In all cases, the differences are less than 20%.
 185 Table s2. Comparison of TOA SW radiative forcing using the simple and Socol models.

Injection rate [Mt-s yr ⁻¹]	Injection type	Injection location	Flux change simple model [W m ⁻²]	Flux change SOCOL [W/m ⁻²]
2	SO ₂	Equator, 20 km	-0.7	-0.9
2	SO ₂	30°S-30°N, 20-25 km	-0.9	-0.9
2	95 nm particles	30°S-30°N, 20-25 km	-1.4	-1.3
5	SO ₂	Equator, 20 km	-1.2	-1.2
5	SO ₂	30°S-30°N, 20-25 km	-1.7	-1.7
5	95 nm particles	30°S-30°N, 20-25 km	-2.7	-2.8
10	SO ₂	Equator, 20 km	-1.8	-1.9

186

187 **H₂SO₄-injection methods**

188 Carrying payloads for deployment between 20 and 25 km altitude via aircraft is non-trivial. An
189 aircraft engineering study of geoengineering by Aurora Flight Sciences is currently investigating the
190 details of this task, thus we will not address it in this work. There are two plausible routes to generate
191 vapor-phase H_2SO_4 on an aircraft. The most direct method is to transport liquid H_2SO_4 to the
192 stratosphere directly and vaporize it onboard the aircraft. The power required for vaporization is not
193 trivial, but is small compared to the aircraft engines thermal power rating. Assuming an aircraft
194 velocity of 240 m sec^{-1} (0.8 Mach) and an injection rate of $100 \text{ kg H}_2\text{SO}_4 \text{ km}^{-1}$, the power is 12 MW.

195 The potential alternative method is to transport elemental sulfur and convert it to H_2SO_4 in situ.
196 The advantage of this method, if feasible, is that the mass of S is only 32.6% the mass of H_2SO_4 , so
197 potentially much less mass must be lifted to the stratosphere compared to lifting H_2SO_4 directly. This
198 conversion process is one of the largest (by mass) processes in industrial chemistry [*Kirk-Othmer*,
199 2004]. The standard process involves combustion of S in air producing SO_2 that is converted to SO_3 in
200 the presence of a vanadium catalyst [*Kirk-Othmer*, 2004]. SO_3 must then combine with water vapor to
201 form H_2SO_4 . Air for the process would be supplied by turbine bleed air; preliminary estimates from an
202 ongoing aircraft engineering study of geoengineering by Aurora Flight Sciences suggest that sufficient
203 air could be available. The H_2O flux from fuel combustion will be larger than that required to form
204 H_2SO_4 under most plausible injection scenarios. The timescale for recombination will likely not affect
205 the results because the size distribution of particles generated in the aircraft plumes was found not to be
206 strongly dependent on nucleation and condensation rates. However, a significant amount of
207 engineering would be required to determine if the necessary amounts of sulfur could be burned using
208 this method on an aircraft. Lightweight systems would need to be developed to make sulfuric acid,
209 Preliminary discussions with NORAM engineering, a firm that specializes in design of sulfuric acid
210 plants, suggests that existing designs would not work for this purpose, but that designs using specialty
211 catalysts might be feasible, however we have not explored this approach in any detail.

212

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